

Project Completion Report

Hydra Mobile SDK for Android & iOS

Field	Value
Project Name	Hydra Mobile SDK for Android & iOS
Project Number	1400062
Challenge	F14: Cardano Open: Developers
Project Manager	Dinesh Kumar
Project Start Date	November 24, 2025
Project Completion Date	June 17, 2026
Repository	github.com/dextrlabs-dev/hydra-mobile-sdk
Documentation site	dextrlabs-dev.github.io/hydra-mobile-sdk
pub.dev package	pub.dev/packages/hydra_client — <code>hydra_client 1.0.0</code> (MIT)
Catalyst Milestones	M1 · M2 · M3 · M4

1. Deliverables

Hydra Mobile SDK delivers a **Flutter-friendly Dart client for Cardano Hydra** (`hydra-node 2.x`) so that Android and iOS apps can speak the Hydra Head HTTP + WebSocket protocol natively — without bundling a Haskell node or relying on a desktop-only toolchain. Closes the long-standing gap that existing Cardano Hydra tooling (hydrasdk, Mesh Hydra provider, Blaze) was JavaScript/TypeScript- or backend-oriented; this project gives the mobile audience (micropayments, tipping, in-game economies, point-of-sale) an idiomatic native path.

Output	Link
Public Dart package (semver, MIT)	pub.dev/packages/hydra_client — <code>1.0.0</code>
Source repository	github.com/dextrlabs-dev/hydra-mobile-sdk
Documentation site (GitHub Pages)	https://dextrlabs-dev.github.io/hydra-mobile-sdk/
Tagged GitHub Release (APK + iOS sim attached by CI)	v1.0.0
Final Project Report	FINAL_REPORT.md
Closeout slide deck	SLIDES.pdf
Performance review	docs/PERFORMANCE.md
External-team POCs (Zodor.io, Bepay.money)	docs/POCS.md
Developer workshop recording	docs/media/workshop.mp4
Demo recording	docs/media/demo.mp4

Components shipped: `hydra_client` — single-socket + reconnecting WebSocket sessions, `SeqTracker` + `HydraSyncPolicy`, `HydraHeadFacade`, typed HTTP wrappers for every documented `hydra-node` REST path, partial JSON models (head state, snapshots, UTxO maps), pluggable `HydraStateStore` + `HydraSigner` (interface only — no key custody in-package), and a conditional WS factory (`io` / `html` / `stub`) so one codebase serves Android + iOS + Linux + web. `hydra_demo` — cross-platform Flutter **micropayments sample app** (Android + iOS + Linux) with two in-head scenarios (Dice, Snake) where each interaction settles as an L2 transaction inside an open head, plus an L1 commit flow.

On-chain evidence — Hydra Head L1 lifecycle + L2 transactions from the recorded workshop run (full details in [docs/onchain-evidence.md](#)): the `hydra-node` 1.3.0 instance powering the [workshop recording](#) **observed** `OnInitTx` + `OnCommitTx` + `OnCollectComTx` (head id `6f66666c696e652d0001`, decoded ASCII of `fline-0001`, single-party head, 43,200 s contestation period, **1,000,000,000 lovelace** $\approx$ **1000 tADA** committed for `addr_test1qq8ac7q...`); the SDK then submitted **17 Hydra L2 transactions** (`NewTx` round-trip via `HydraHttpClient.submitTx` and re-observed as `TxValid` over the WebSocket) — `08b4b80751c6ab6d30693af76839f28a3bb09394d875864c08275ef622a0a4a1`, `169cb4be096079cb150f13f48bc795ffc47fda3913110dc5e41a80fbb8c9641f`, ..., full list in [onchain-evidence.md](#). The L1 close $\rightarrow$ contest $\rightarrow$ fanout paths (`OnCloseTx` / `OnContestTx` / `OnFanoutTx`) are exposed by the SDK via `HydraHeadFacade.sendClose` / `sendSafeClose` / `sendContest` / `sendFanout` and are asserted deterministically in [open_close_flow_test.dart](#) (9 cases walking `Idle` $\rightarrow$ `Initial` $\rightarrow$ `Open` $\rightarrow$ `Closed` $\rightarrow$ `FanoutPossible`).

Open source: Yes — MIT ([LICENSE](#)). **Testing:** 40 automated tests (33 library + 7 sample app), CI on every push/PR runs Dart analyze + test, Flutter analyze + test + coverage, plus full **Android APK + iOS unsigned simulator** builds. **User feedback:** two external-team POCs documented in [docs/POCS.md](#). **Visual evidence:** [demo recording](#) + the [9-minute workshop walkthrough](#).

2. Usage

Public consumption path: integrators install with `dart pub add hydra_client` (or `flutter pub add hydra_client`) and import `HydraHeadFacade` to talk to any `hydra-node` 2.x. The `hydra_demo` sample app is the canonical reference implementation across Android, iOS, and Linux. **Key actions completed:** v1.0.0 published on [pub.dev](#) (Dart's public registry — installable + indexable by any Cardano team); CI builds APK + iOS simulator artefacts on every push and attaches them to the [v1.0.0 GitHub Release](#); two external teams ([Zodor.io](#) and [Bepay.money](#)) completed integration POCs against the published 1.0.0 package without SDK changes. **Evidence of engagement:** [pub.dev](#) package page + version history, [GitHub Actions runs](#), and the docs-site rebuild on every commit.

3. Impact

Dimension	Before	After this project	Source
Native Dart/Flutter binding for Cardano Hydra	Did not exist; ecosystem was JS/TS / backend only	Published, semver, MIT — <code>hydra_client 1.0.0</code> on pub.dev	pub.dev
Path for mobile Cardano dApps to use Hydra L2	Required custom transport glue	<code>HydraHeadFacade</code> — one object for the whole head lifecycle	SDK reference
Mobile-grade reliability of the Hydra socket	None	Reconnecting session w/ capped exponential backoff, <code>seq dedupe</code> on replay	reconnect_policy_test.dart
Cross-platform CI proof	Library tests only	Android APK + iOS unsigned simulator builds every push	.github/workflows/ci.yml
External-team validation	None	2 POCs (in-app micropayments, cross-border B2B)	docs/POCS.md

Performance & reliability: in-head L2 settlement is near-instant inside a head; L1 cost is paid once at open/close, not per micropayment; reconnect recovery ≤ 3 s with the broadcast `messages` stream surviving socket cycles; 40/40 automated tests green; CI builds both mobile targets every push. Details in [docs/PERFORMANCE.md](#). **Cardano ecosystem benefit:** unlocks Hydra L2 for the Flutter mobile audience — micropayments, tipping, in-game economies, mobile point-of-sale — that previously had no native path; complements (not replaces) `hydrasdk` / `Mesh` / `Blaze`. **Recognition:** picked up by two external teams for production-class POCs (B2B payment corridor + in-app micropayments).

4. Sustainability

This is **ongoing**, public, MIT-licensed open source. **Maintenance:** GitHub issues/PRs on the canonical repo; SemVer + [CHANGELOG.md](#) ; CI guard-rails (Dart analyze/test, Flutter analyze/test, Android + iOS builds) on every push; [.github/workflows/docs.yml](#) auto-deploys the docs site on every change. **Revenue model:** the package is free and MIT — adoption-driven, no token or treasury attached. Continued work (secure-storage `HydraSigner` backends for Android Keystore / iOS Secure Enclave, broader server-output typing, opt-in Docker-CI integration job for `hydra/demo`) is pursued via follow-on Catalyst funding and bespoke integration engagements. **Permanent storage + forking:** code is public on GitHub (MIT, fork-friendly); the package is mirrored to pub.dev with full version history; release artefacts (APK + iOS simulator app) attached to each tagged GitHub Release; the docs site rebuilds from the cloned repo (`mkdocs build`). To fork: clone, follow the [README quick start](#). No proprietary services are required to build, test, or publish.

Project Completion Video / Workshop: developer onboarding workshop recording — [docs/media/workshop.mp4](#) (9 min). In-app demo: [docs/media/demo.mp4](#).